

REPORT ON
Micromachined Gas Sensors- Design and Simulations
Training Seminar(EEN-499)

MITACS Globalink Research Internship
Supervisor Professor : Dr Arezoo Emadi

Submitted by
Bhaskar Garg
Electrical Engineering Department, IIT Roorkee
bgarg@ee.iitr.ac.in



DEPARTMENT OF Electrical and Computer Engineering
University of Windsor
Windsor, ON N9B 3P4, Canada

TABLE OF CONTENT

Chapter 1

1.1 INTRODUCTION	3
1.2 MOTIVATION	3
1.3 OBJECTIVES	4

Chapter 2

2.1 TECHNICAL DETAILS	5
2.2 SENSOR DESIGN AND SIMULATIONS	6
2.2.1. Design 2 Results	6
2.2.2. Design 3 Results	8
2.2.3. Design 6 Results	9
2.2.4. Design 7 Results	11

REFERENCES	14
-------------------	-----------

Chapter 1

1.1 INTRODUCTION

In any system if we need to measure anything quantitatively, we need a sensor. Sensor measures a physical quantity and converts it into a signal that be measured by any observer or any instrument. In electrical systems, it is convenient to convert the measured signal into electrical parameters like resistance and capacitance.

On change of physical/environmental factors the resistance or capacitance of the measuring sensor changes. The change in electrical quantity can be observed using micro controllers or processors. In this process, it becomes very important to know the sensitivity of the sensor. This will help us understand the change in electrical quantity with respect to the physical quantity i.e. on per unit change of physical quantity(that we are sensing) how much change in electrical quantity is expected.

Instead of directly making these sensor designs and testing them on field, we can simulate the designs on COMSOL and get the exact sensitivity and results of the designs. In this way various designs can be tested. Their capacitiva and resistive results can be deduced based on the changing physical parameters.

1.2 MOTIVATION

Various different designs of Sensors can be imagined to prepare a working gas sensor. Choosing the aproprate parameters for the sensors become necessary to get desired results. We will be performing several finite mesh element analysis simulations on various sensor designs to check them on their sensitivity. We will try to look how the electrical parameters(Capacitance or Resistance) changes with changing environmental factors.

Obtaining all the desired capacitance and resistances of different sensor designs will give a idea on their field performance. The simulation results give clarity on the choice of sensor designs for various applications. Simulations will make us aware of any glitch or error in the sensor designs. The error can be rectified in design stage which would be better than encountering it after fabrication.[1-4]

1.3 OBJECTIVES

- Simulations of Design 2 and Design 3. The COMSOL files for designs are provided and capacitance of these designs need to be measured on changing electrode count and area of sensor.
- Simulating Capacitance and resistance values for spiral Design 6 sensor, with changing physical parameters and spiral count.
- Preparing the sensor Design 7 from scratch in COMSOL. The design 7 built from very basics and learning to the desirable geometry using basic building block functions in COMSOL.
- Simulation of various design 7 parameters after the design file is ready.

Chapter 2

2.1 TECHNICAL DETAILS

Reactance of sensor can be mathematically calculated using the formulas shown in the figure. As the adsorption occurs the properties of the material changes and change in reactance/capacitance can be seen. Secondly the reactance changes due to the geometry of the sensor design, like Area and diameter, these parameters plays an important role in getting the sensitivity as these factors play crucial role in determining the slope of the capacitance curve.[5-7]

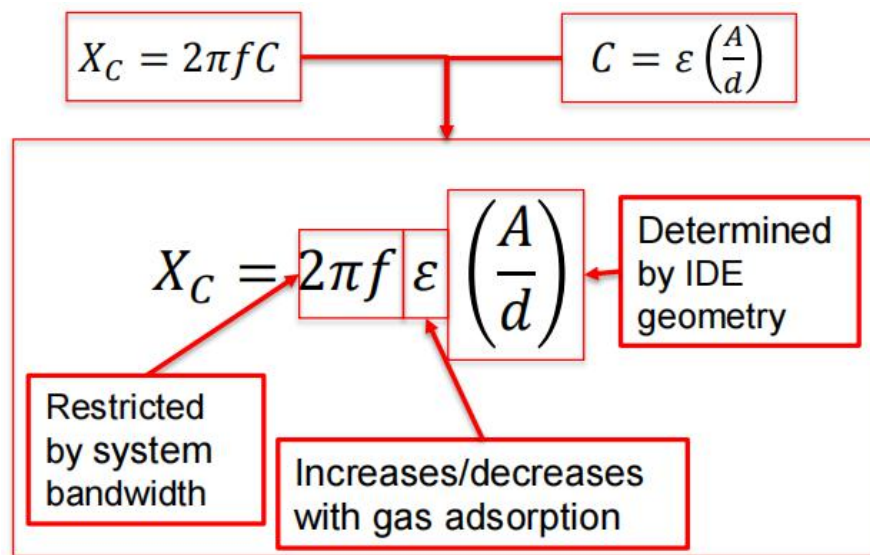


Figure 1.1: Formula for Capacitive impedance calculations using basic physical parameters

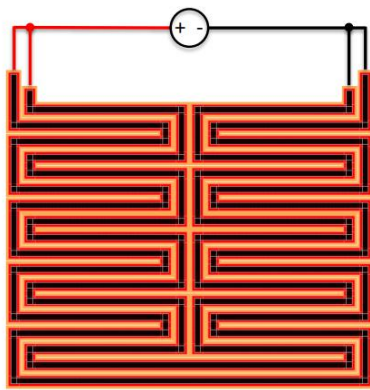
It is important to notice that Area(A) and Diameter(D) are not the only factors to determine slope or sensitivity[as in Figure1.1]. In practical conditions, there are some fringing fields as well, which don't exactly lie between the capacitor plates but contribute to capacitance. That's why it becomes necessary to perform simulation on COMSOL instead of having an idea of slope mathematically calculating Area and Diameter.[8-9]

2.2 SENSOR DESIGN AND SIMULATIONS

Design 2, 3, 6, 7 were simulated using mesh refinement in COMSOL. The results for capacitance and resistance was recorded for change in different parameters. These can be further used to check the sensor designs on various parameters.

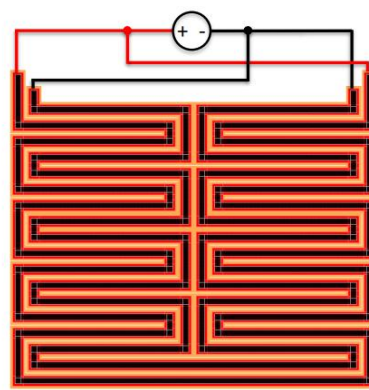
2.2.1. Design 2 Results

The design 2 is a hybrid microheater/IDE as shown in Figure 2.2. The respective configurations can be changed by just switching the power source.



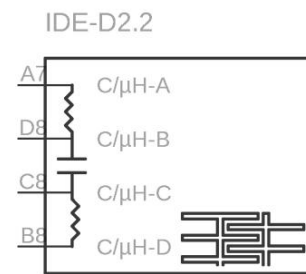
Configured as two microheaters

Figure 2.1a



Configured as IDE

Figure 2.1b



Circuit Model for Design 2

Figure 2.1c

Figure 2.1: Design of Capacitive sensors as Microheater(a) and IDEs(b). The placement of these capacitors in the real time circuit in series(c).

The capacitive simulations for the design are as in the Table 2.1 and Figure 2.2.

%RH	Relative Permittivity	Sens Material Thickness	n (Electrode Column Pairs)	Length	Width	Area	Air Side Gap	Air Top Gap	Air Bottom Gap	Mesh Refinement	Capacitance (F)
0%	3.05	25 μm	1	668 μm	172 μm	114896 μm^2	100 μm	100 μm	0 μm	Extremely Fine	9.77E-15 F
			2	668 μm	288 μm	192384 μm^2	100 μm	100 μm	0 μm	Extremely Fine	1.57E-14 F
			3	668 μm	404 μm	269872 μm^2	100 μm	100 μm	0 μm	Extremely Fine	2.17E-14 F
			4	668 μm	520 μm	347360 μm^2	100 μm	100 μm	0 μm	Extremely Fine	2.77E-14 F
			5	668 μm	636 μm	424848 μm^2	100 μm	100 μm	0 μm	Extremely Fine	3.37E-14 F

Table 2.1: Results for capacitive simulations of sensor design.

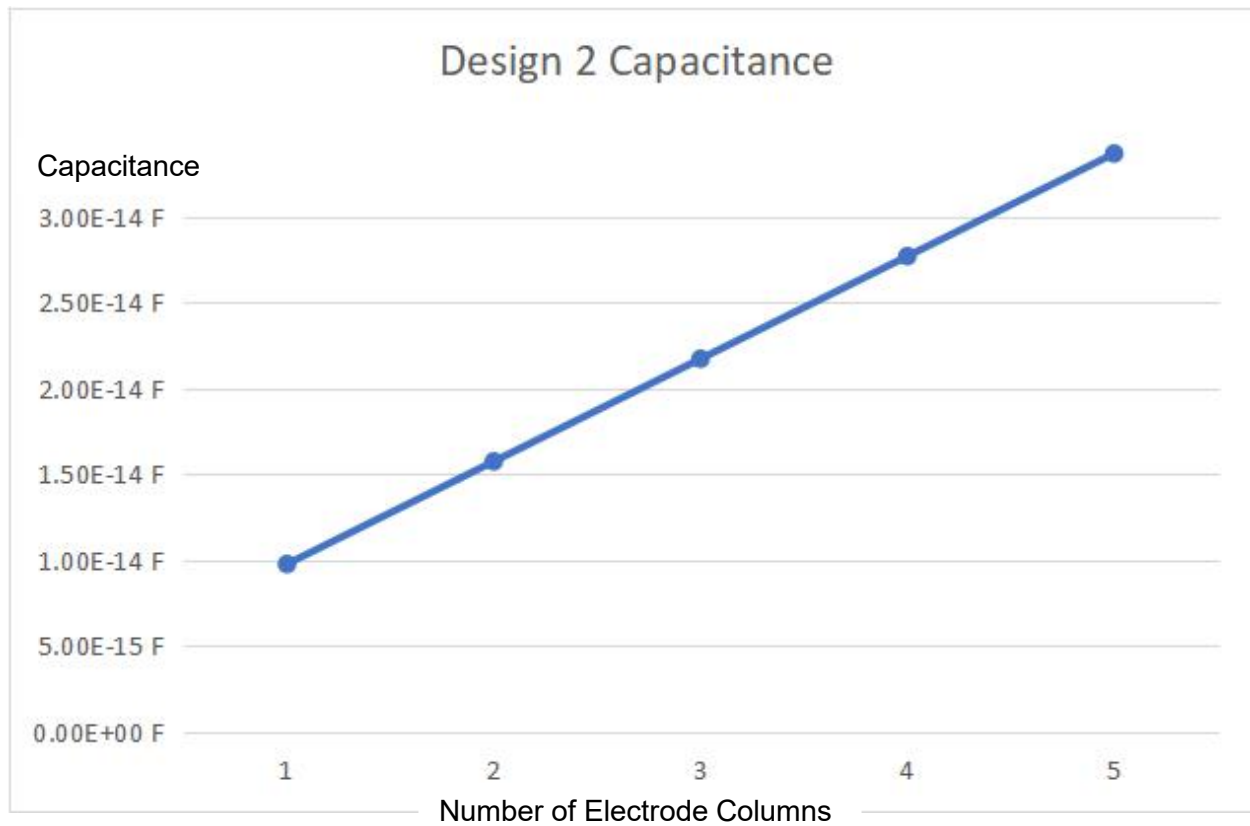


Figure 2.2 Graph to show the variation of capacitance of this design with respect to change in number of electrode columns in the design.

The fringing field capacitance was calculated for this design in 1 electrode case to be 4.72E-15 F.

2.2.2. Design 3 Results

The design 3 utilizes all available layout area while offering the user flexibility. Different regions of the layout area can be isolated as needed as seen in Figure 2.3.

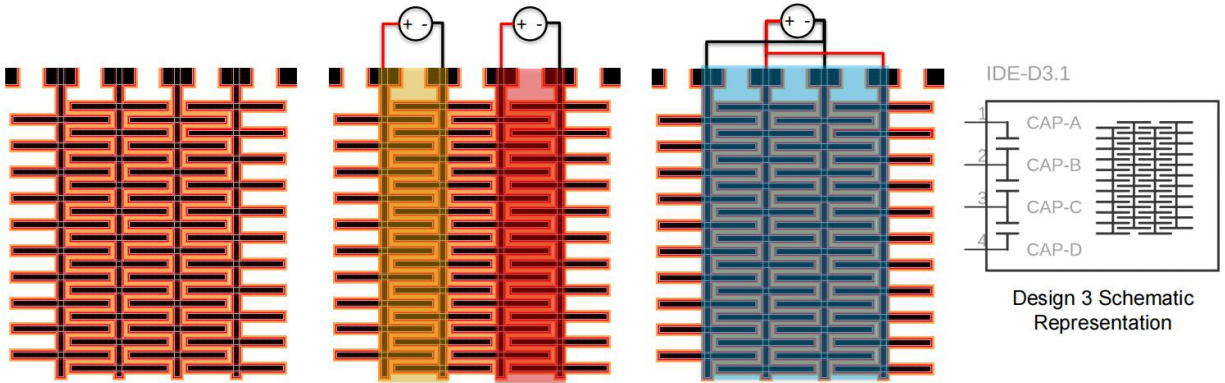


Figure 2.3a

Figure 2.3b

Figure 2.3c

Figure 2.3d

Figure 2.3: The basic design layout of the design 3 in 2.3a. The demonstration of isolation of rows in 2.3b and 2.3c. The placement of capacitor in circuits in 2.3d.

The capacitive simulation results for the Design 3 are in the table 2.2 and Figure 2.4.

%RH	Relative Permittivity	Sens Material Thickness	n_rows	n_columns	Length	Width	Area	Air Side Gap	Air Top Gap	Air Bottom Gap	Mesh Refinement	Capacitance (F)
0%	3.05	25 um	1	5	1495.5 um	56 um	83748 um ²	100 um	100 um	0 um	Extremely Fine	8.01E-14 F
			2	5	1495.5 um	114 um	170487 um ²	100 um	100 um	0 um	Extremely Fine	1.99E-13 F
			3	5	1495.5 um	172 um	257226 um ²	100 um	100 um	0 um	Extremely Fine	3.18E-13 F
			4	5	1495.5 um	230 um	343965 um ²	100 um	100 um	0 um	Extremely Fine	4.37E-13 F
			5	5	1495.5 um	288 um	430704 um ²	100 um	100 um	0 um	Extremely Fine	5.56E-13 F

Table 2.2: Results for capacitive simulations of sensor design.

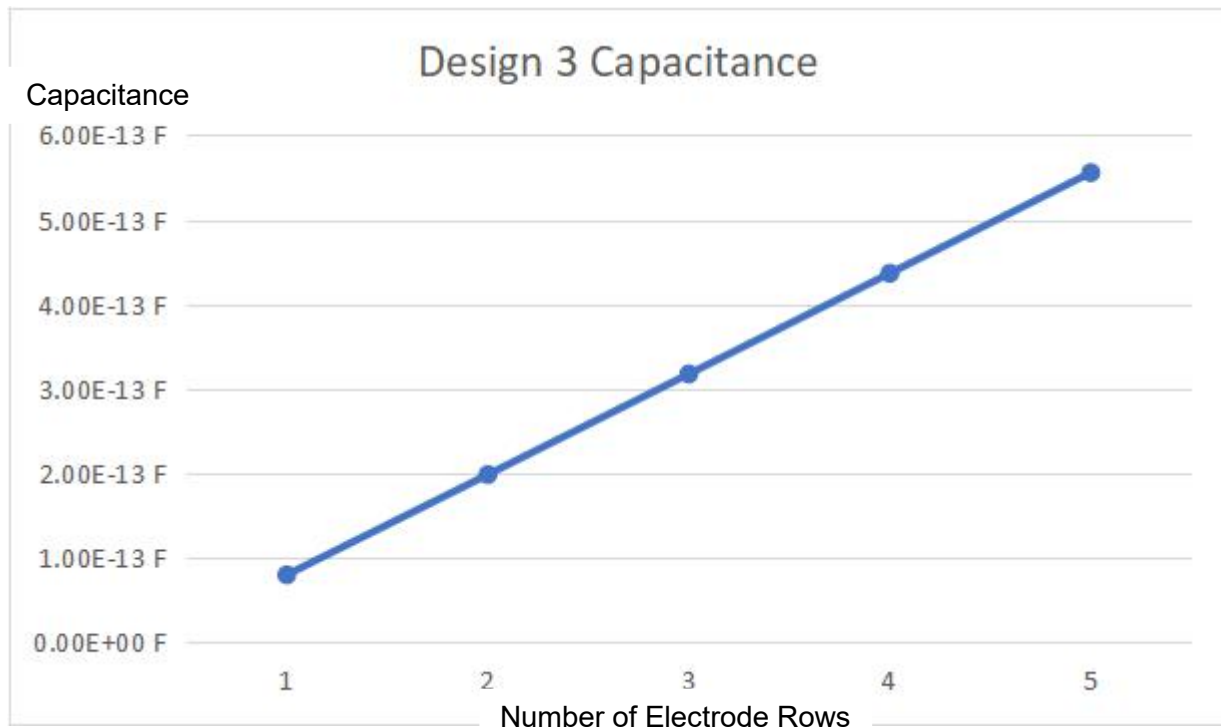
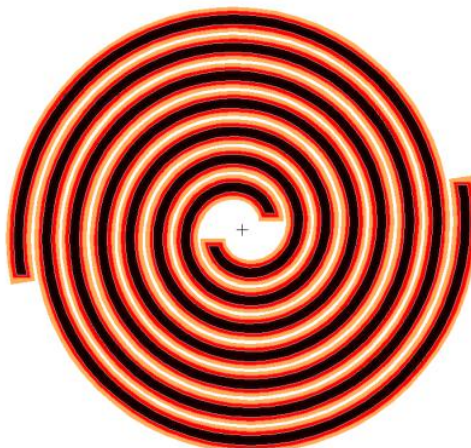


Figure 2.4 Graph to show the variation of capacitance of this design with respect to change in number of electrode rows in the design.

The fringing field capacitance was calculated for this design in 1 electrode case to be $5.07\text{E-}14$ F.

2.2.3. Design 6 Results

This is a spiral design of capacitor as in Figure 2.5.



Design 6

Figure 2.5: The sensor design of spiral shaped capacitor.

The capacitive simulation results for the Design 6 are in the table 2.3 and Figure 2.6.

%RH	Relative Permittivity	Sens Material Thickness	Turns	Length	Width	Area	Air Side Gap	Air Top Gap	Air Bottom Gap	Mesh Refinement	Capacitance (F)
0%	3.05	25 um	1			33644 um2	100 um	100 um	0 um	Extremely Fine	1.93E-14
			2			83957 um2	100 um	100 um	0 um	Extremely Fine	6.94E-14
			3			156880 um2	100 um	100 um	0 um	Extremely Fine	1.47E-13
			4			252420 um2	100 um	100 um	0 um	Extremely Fine	2.52E-13
			5			370580 um2	100 um	100 um	0 um	Extremely Fine	3.84E-13

Table 2.3: Results for capacitive simulations of sensor design.

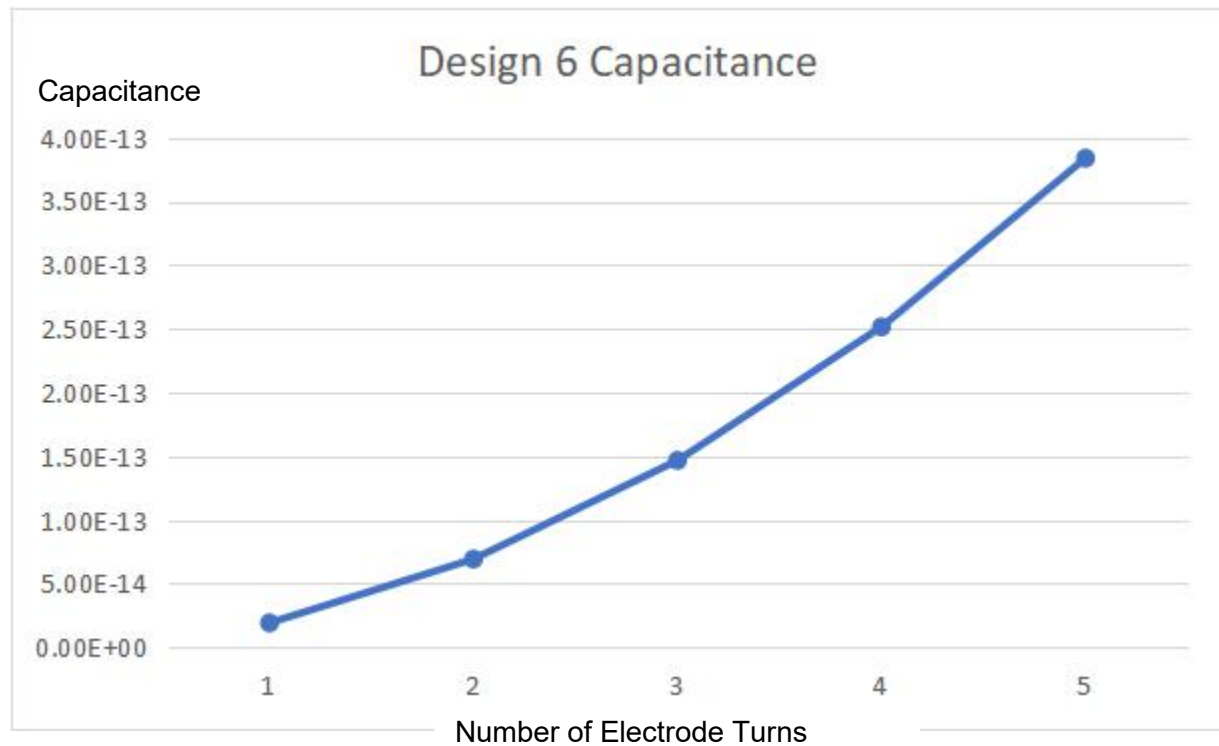


Figure 2.6 Graph to show the variation of capacitance of this design with respect to change in number of electrode rows in the design.

The length and breadth column of the table are left empty as the design is spiral shaped so we directly got the area with respect to number of turns.

2.2.4. Design 7 Results

Design 7 was built in COMSOL as an array of nitride breached area etched within a grid of polysilicon as shown in Figure 2.7. In the design polysilicon grid will act as top electrode and chip substrates will act as the bottom electrode. As we are taking substrate as one of the electrodes, this inclusion may decrease the overall parasitic capacitance for this particular design.[10, 11]

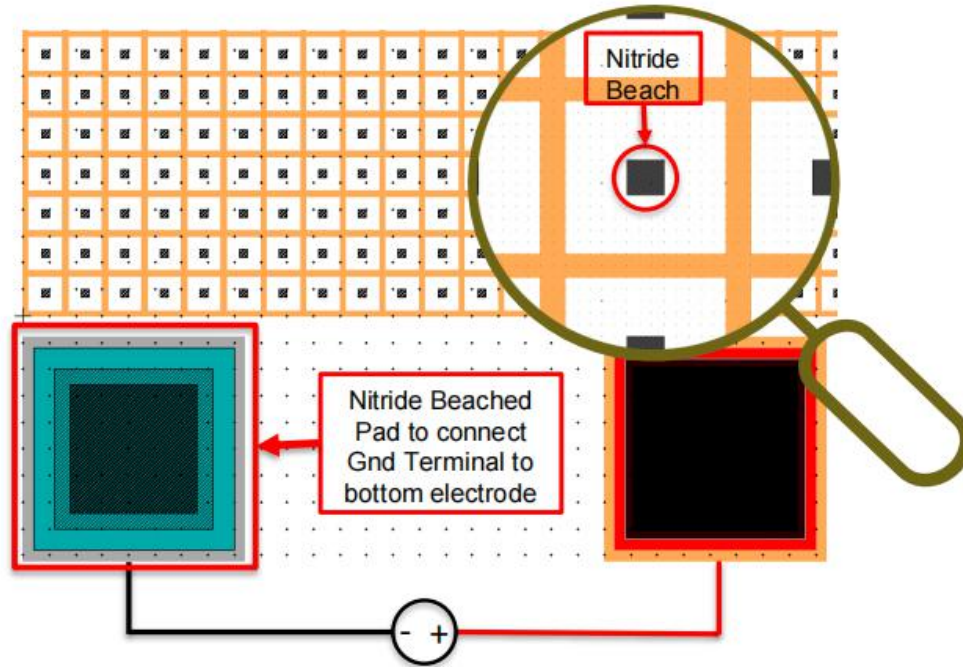


Figure 2.7 Detailed design overview on Design 7.

The capacitance values for Design 7 for different electrode counts is shown in Table 2.4 and Figure 2.8.

%RH	Relative Permittivity	Sens. Material Thickness	n_rows	n_columns	Length	Width	Area	Air Side Gap	Air Top Gap	Air Bottom Gap	Mesh Refinement	Capacitance (F)
0%	3.05	25 um	20	20	603 um	603 um	363609 um ²	100 um	100 um	0 um	Extremely Fine	1.36E-11 F
			22	22	663 um	663 um	439569 um ²	100 um	100 um	0 um	Extremely Fine	1.64E-11 F
			24	24	723 um	723 um	522729 um ²	100 um	100 um	0 um	Extremely Fine	1.95E-11 F
			26	26	783 um	783 um	613089 um ²	100 um	100 um	0 um	Extremely Fine	2.28E-11 F
			28	28	843 um	843 um	710649 um ²	100 um	100 um	0 um	Extremely Fine	2.64E-11 F
			30	30	903 um	903 um	815409 um ²	100 um	100 um	0 um	Extremely Fine	3.03E-11 F

Table 2.4: Results for capacitive simulations of sensor design.

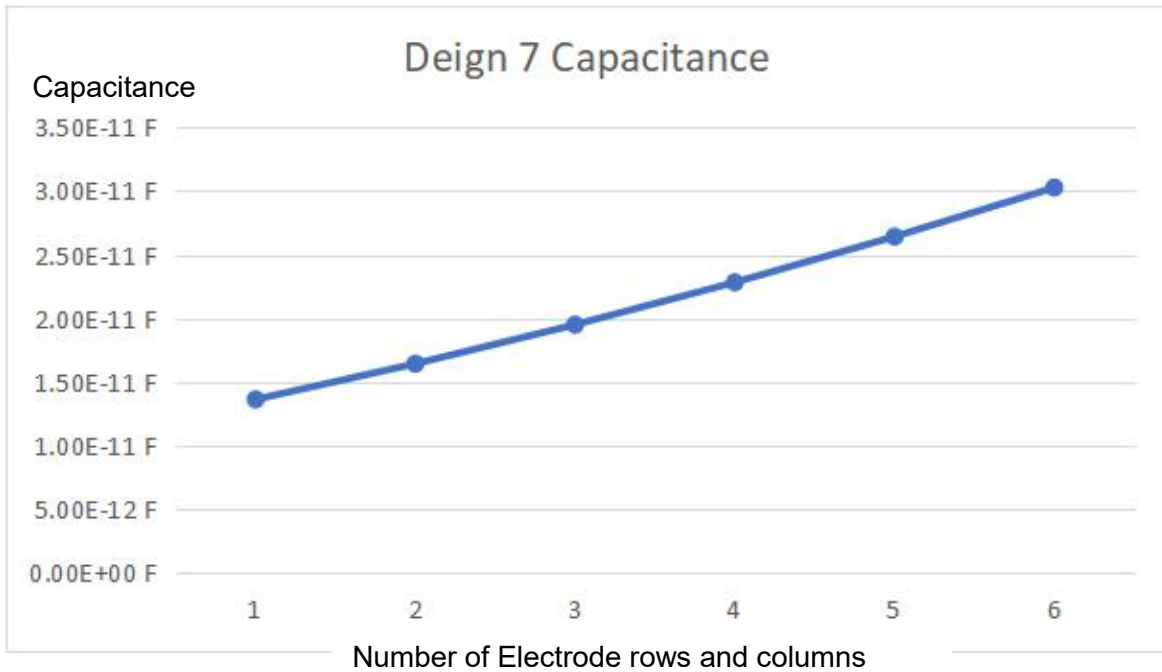


Figure 2.8 Graph to show the variation of capacitance of this design with respect to change in number of electrode rows in the design.

The resistance values for Design 7 is shown for various electrode values in Table 2.5 and Figure 2.9.

%RH	Resistivity	Sens. Material Thickness	n1	n2	Length	Width	Area			Mesh Refinement	Resistance
0%	500	120 um	6	6	183 um	183 um	33489 um2			Extremely Fine	3002800 Ω
			8	8	243 um	243 um	59049 um2			Extremely Fine	1689600 Ω
			10	10	303 um	303 um	91809 um2			Extremely Fine	1081600 Ω
			12	12	363 um	363 um	131769 um2			Extremely Fine	751230 Ω
			14	14	423 um	423 um	178929 um2			Extremely Fine	552010 Ω

Table 2.5: Results for resistance simulations of sensor design.

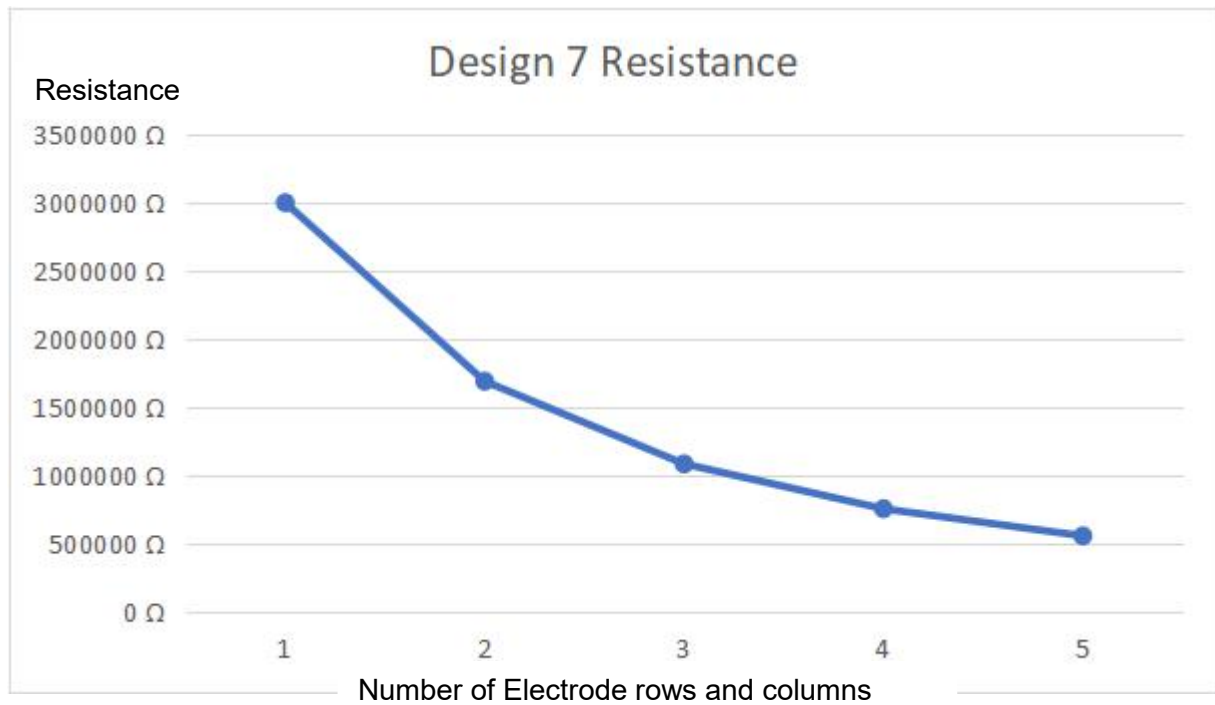


Figure 2.9 Graph to show the variation of resistance of this design with respect to change in number of electrode rows in the design.

Using these values and values of capacitance/resistance at different Relative Humidity levels, the sensitivity for particular number of electrodes can be calculated. Which will give us the optimum number of electrodes for required sensitivity.

REFERENCES

- [1] A. D'Amico *et al.*, "An investigation on electronic nose diagnosis of lung cancer," *Lung Cancer*, vol. 68, no. 2, pp. 170–176, May 2010.
- [2] R. Gasparri *et al.*, "Volatile signature for the early diagnosis of lung cancer," *J. Breath Res.*, vol. 10, no. 1, p. 016007, Feb. 2016.
- [3] X. Chen *et al.*, "A study of an electronic nose for detection of lung cancer based on a virtual SAW gas sensors array..." *Meas. Sci. Technol.*, 2005.
- [4] R. F. Machado *et al.*, "Detection of lung cancer by sensor array analyses of exhaled breath," *Am. J. Respir. Crit. Care Med.*, vol. 171, no. 11, pp. 1286–1291, Jun. 2005.
- [5] H. G. Moon *et al.*, "Chemiresistive Electronic Nose toward Detection of Biomarkers in Exhaled Breath," *ACS Appl. Mater. Interfaces*, Aug. 2016.
- [6] S. R. Chowdhury *et al.*, "Circuit arrangement to suppress crosstalk in chemo-resistive sensor arrays," *IET Sci. Meas. Technol.*, vol. 12, no. 8, Nov. 2018.
- [7] U. Simon *et al.*, "Design Strategies for Multielectrode Arrays Applicable for High-Throughput Impedance..." *J. Comb. Chem.*, Sep. 2002.
- [8] J. Wu, L. Wang, and J. Li, "Design and Crosstalk Error Analysis of the Circuit for the 2-D Networked Resistive Sensor Array," *IEEE Sens. J.*, 2015.
- [9] J. A. H. López, Ó. Oballe-Peinado, and J. A. Sánchez-Durán, "A Proposal to Eliminate the Impact of Crosstalk on Resistive Sensor Array Readouts," *IEEE Sens. J.*, 2020.
- [10] B. Mondal *et al.*, "Fabrication and packaging of MEMS based platform for hydrogen sensor using ZnO–SnO₂ composites," *Microsyst. Technol.*, Dec. 2016.
- [11] A. Cowen, B. Hardy, R. Mahadevan, and S. Wilcenski, "PolyMUMPs Design Handbook a MUMPs® process," *MEMSCAP Inc*, 2013.